

APOLLO 10	
LAUNCH - LOI CHECKLIST	
PART NO.	S/N
SKB32100079-303	1001

POST-INSERTION CHECKLIST

CDR 1 GMBL MTRS (4) - OFF (LMP confirm)
TVC SERVO PWR (2) - OFF
MN BUS TIE (2) - OFF (LMP)
SECS PYRO ARM (2) - SAFE
SECS LOGIC (2) - OFF
cb SECS ARM (2) - open
ELS - MAN (verify)
CM RCS LOGIC - OFF
CAB PRESS REL vlv (2) - NORMAL/LATCHED
PLSS 02 vlv - OFF
cb FLOAT BAG (3) - open
cb ELS BAT (2) - open
cb PL VENT - open
DIRECT 02 vlv - CLOSED
ROT CONTR PWR DIRECT - OFF
EMS FUNC - OFF

VAN LOS
(00:15:32)

CYI AOS
(00:16:29)

CMP 2

MONITOR LV TANK PRESS
AP < 36 psid (OXID > FUEL)
AP < 26 psid (FUEL > OXID)
LOX TK PRESS > 50 psia
* EMERGENCY CSM/LV SEP pg L/5-3*
SM RCS HTRS (4) - PRIM
CM RCS PRPLNT (2) - OFF, tb (2) - bp
C/W - NORMAL
BPC JETT KNOB - 180° from BPC JETT

CAUTION

If the BPC JETT Knob can not be positioned 180° away from the BPC JETT position before opening the side hatch, a pin in the plunger may be sheared possibly allowing the pressure seal to be broken. Cabin repressurization may not be possible due to overboard venting around the BPC plunger.

LMP

- GN2 VLV HNDL - pull
- HATCH GEAR BOX - LATCH (verify)
- ACTR HNDL SELECTOR - Neutral
- STRUT UNLOCK LANYARD (2) - stow
- 3 cb WASTE H2O/URINE DUMP HTRS (2) - close
- FC REACS vlv - NORM
- H2 PURGE LINE HTR - ON

SYS MONITORING & CHECKING

- 1 SM RCS Ck (CMP) pg S/1-1
- 2 CM RCS Ck (CMP) pg S/1-1
- 3 C & W Ck (CMP) pg S/1-17
- 4 CMP to LEB for MN REG Ck pg S/1-6.
- 5 *EMER CAB PRESS* → SUIT CKT RET vlv - OPEN (pulled) (CMP)
vlv - BOTH (verify) Remove Helmets & Gloves
cb DIRECT ULLAGE (2) - open (CDR)
- 6 SEC RAD LEAK Ck
Monitor SEC ACCUM QUANTITY (LMP)
SEC GLY To RAD vlv - NORM for (30 sec)
then BYPASS (CMP)

- +20:00 7 ECS Post Insertion Config (CDR)
- GLY RSVR BYPASS vlv - OPEN
 - GLY RSVR OUT vlv - CLOSE
 - GLY RSVR IN vlv - CLOSE
 - ECS RAD FLOW CONT - PWR
 - PRIM GLY TO RAD vlv - NORMAL (push)
 - LMP note PRIM ACCUM QTY
 - ECS RAD HTR - PRIM 1 (LMP)

ECS RAD TEMP PRIM OUT below PRIM IN

- *If outlet temp after 5 min*
- *above INLET TEMP *
- *PRIM GLY TO RAD vlv - *
- * BYPASS (pull) *
- *Recheck in 10 min *

ECS RAD tb - gray
GLY EVAP TEMP IN - AUTO

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POST INSERTION

- 8 DRINKING WATER SUPPLY vlv - ON
POT H2O HTR - MNA

CYI LOS
(00:23:36)

PCM BIT RATE - LOW
UP TLM CMD - RESET
VHF AM A - SIMPLEX
VHF AM B - off (ctr)

- 9 FC Purge Ck (LMP)
H2/O2 PURGE (6) - ON (mom)
Observe Flow rate inc
Reset MA (as req'd)
H2 PURGE LINE HTR - OFF

PRE-TLI SYS VERIFICATION & MONITORING

- 1 EPS Mon Ck (LMP) pg S/P-2
- 2 ECS Mon Ck pg S/P-3
- 3 SPS Monit Ck (LMP) pg S/1-1
- 4 GDC ALIGN (CDR) pg G/1-87 (Record Drift) _____
BMAG MODE (3) - RATE 2
Install COAS
cb COAS TUNL LTG MNB (Pnl 226) - closed
- 5 UNSTOW SEQ CAMERA BRACKET & ORDEAL (CDR)
- 6 MOUNT ORDEAL BOX & Initialize
- 7 ECS REDUNDANT COMPONENT Ck less MN REG
Ck (LMP) pg S/1-6
- 8 UNSTOW TV CAMERA & ACCESSORIES
- 8A UNSTOW & CONFIGURE CAMERAS (CMP)
Set up Data Acq. Camera/18mm lens/CEX 368
Film/mirror
Settings (f11, 250, ∞) 6 fps, .7 mag/
Mount RH Rndz Window ~~250, 250, 250~~
Electric Hasselblad/80mm lens/
CEX 368 film/Spotmeter (S, 250, focus)
20 frames

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SUNSET 9 OPTICS DUST COVER JETT (CMP)
(00:31:33)

TAN AOS
(00:37:01)

10 MCCH - G/N STATUS
Z Torquing Angle _____

TAN LOS
(00:42:35)

11 IMU REFSMMAT Realign Check (P52), (CMP)
If IMU is realigned,
Realign GDC (CDR)

CRO AOS OOE
(00:52:15) RETICLE BRIGHTNESS - DIM
CRO LOS Stow Optics Eyepieces
(00:58:01)

HSK AOS(S) 12 Two-way USB VOICE Ck
(00:59:30) Report Gyro Torquing Angles

HSK LOS
(01:05:00)

SUNRISE
(01:09:02)

HTV AOS
(01:10:00)

HTV LOS
(01:13:00)

US AOS
(01:28:15)

13 SCS ATT Ref Comp Ck
V16 N20E
FDAI SELECT - 1
FDAI SOURCE - ATT SET
ATT SET - GDC
ATT SET dials - null FDAI 1 err needles
Key VERB when nulled(freeze display)

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CSM 106

Record from DSKY:

R____, P____, Y____

Record ATT SET dials:

R____, P____, Y____

FDAI SEL - 1/2

14 Extend Docking Probe

cb DOCK PROBE (2) - close (verify)

PROBE EXTD/REL - EXTD/REL until ~~the~~ *full probe extension*

	EXT	RET
FULL EXT	Gray	Gray
FULL RET	bp	bp
PART RET	bp	Gray

15 COPY TLI, TLI ABORT, & P37 PADS

TLI + 10 min Pitch Angle _____

VAN LOS

(01:49:11)

CYI AOS

(01:50:01) 16 SV UPDATES (MCCH)

17 cb SECS ARM (2) - close

SECS LOGIC (2) - on (up)

MSFN confirm GO for PYRO ARM

TLI

X		:	:		X		:	:		TB6p
X	X	X			X	X	X			R
X	X	X			X	X	X			P
X	X	X			X	X	X			Y
X	X	X	:		X	X	X	:		BT
										Δ VC'
+					+					VI
X	X	X			X	X	X			R SEP
X	X	X			X	X	X			P SEP
X	X	X			X	X	X			Y SEP

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L
4-6

P27 UPDATE

PURP	V			V			V		
GET	⊙	⊙		⊙	⊙		⊙	⊙	
304 01	INDEX			INDEX			INDEX		
02									
03									
04									
05									
06									
07									
10									
11									
12									
13									
14									
15									
16									
17									
20									
21									
22									
23									
24									
N34	HRS	X	X	X			X	X	X
	MIN	X	X	X	X		X	X	X
NAV CHECK	SEC	X	X				X	X	
N43	LAT		0					0	
	LONG								
	ALT	+	0				+	0	

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Changed May 3, 1969

CMS 106

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 Changed

CSM 106

L/4-7		P30 MANEUVER	
SET STARS		T L I + 9 0 /	PURPOSE PROP/GUID
		+	WT N47
R ALIGN _____		0 0 .	P _{TRIM} N48
P ALIGN _____		0 0 .	Y _{TRIM}
Y ALIGN _____		+ 0 0	HRS GETI
		+ 0 0 0	MIN N33
		+ 0 .	SEC
ULLAGE _____			ΔV_X N81
_____			ΔV_Y
_____			ΔV_Z
_____		X X X	R
_____		X X X	P
_____		X X X	Y
		+	H _A N44
			H _P
		+	ΔV_T
HORIZON/WINDOW _____		X X X .	BT
_____		X .	ΔV_C
_____		X X X X	SXTS
_____		+	SFT
_____		+	TRN
		X X X	BSS
		X X ☐	SPA
		X X X ☐	SXP
		0 .	LAT N61
			LONG
		+	RTGO EMS
		+	V10
			GET 0.05G

X	•	GETI ΔVT LONG GET 400K
X ☐	•	
	•	
	•	

L/4-8

P30 MANEUVER

SET STARS				T	L	I	+	9	0	PURPOSE
						/				PROP/GUID
				+						WT N47
R ALIGN					0	0				P TRIM N48
P ALIGN					0	0				Y TRIM
Y ALIGN				+	0	0				HRS GETI
				+	0	0	0			MIN N33
				+	0					SEC
ULLAGE										ΔV_X N81
										ΔV_Y
										ΔV_Z
				X	X	X				R
				X	X	X				P
				X	X	X				Y
				+						H _A N44
										H _P
				+						ΔVT
HORIZON/WINDOW				X	X	X				BT
				X						ΔVC
				X	X	X	X			SXTS
				+					0	SFT
				+				0	0	TRN
				X	X	X				BSS
				X	X					SPA
				X	X	X				SXP
					0					LAT N61
										LONG
				+						RTG0 EMS
				+						V10
										GET 0.05G

P37 FOR TLI + 4				GETI
				ΔVT
X				LONG
X				GET 400K

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C 106

LAUNCH EMERGENCY PROCEDURES

RAPID HATCH OPENING

- 1 GEAR BOX SEL - UNLATCH (verify)
ACTR HANDLE rel - push or squeeze
- 2 ACTR HANDLE - operate (until hatch is unlatched)
*If hatch fails to open *
* GN2 RATCHET HNDL - operate *
* GN2 VLV HNDL - unlock and *
* push (outboard) *

FIRE IN CM DURING BOOST

- 1 CABIN FAN (2) - OFF (verify)
- 2 Monitor EPS indicators for excessive current.
Immediately remove power from affected bus.
If in abort modes I or II:
Verify suit compressor on good AC bus
If in abort mode III with affected bus Main A(B)
TVC GMBL DRIVE (2) - 2 (1)
AC INV 1 (2) AC BUS 1 (2) - OFF
AC INV 2 (1) AC BUS 1 (2) - on (up)
- 3 CAB PRESS REL vlv (RH) - DUMP
- 4 ABORT using appropriate mode

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CONTAMINATION IN CM

- 1 Don O2 masks and/or PGA's immediately
- 2 Evaluate contamination level (isolate & correct source of contamination if possible) and proceed with one of the following steps:
 - a. Retain O2 masks or remain in suit and accept contamination level in cabin.

CAUTION

If in PGA's, adjust DIRECT O2 to maintain suit to cabin $\Delta P > +2$ in. H2O

- b. Retain O2 masks and scrub cabin atmosphere through suit loop. If initially suited, establish partially suited or shirtsleeve configuration and don O2 masks.

CAUTION

Change LiOH cartridges after scrub completed.

- c. Retain PGA's or don PGA's
Verify suit integrity (visually)
Perform Cabin Dump
Perform Cabin Repress

CONTAMINATION IN SUIT

- 1 SUIT COMPR 2 - AC1
- 2 SUIT COMPR 1 - OFF
- 3 DIRECT O2 vlv - OPEN for 1 minute
then CLOSE

If condition persists:

- 4 SUIT COMPR 2 - OFF
- 5 DIRECT O2 vlv - CLOSE
- 6 Doff helmet
- 7 Don emergency O2 masks

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LET FAILS TO JETTISON

LEGS CUT/NO MOTOR FIRE (pyro audible)
LES MOTOR FIRE PB - push
NO RESPONSE to ABORT SYS TWR JETT switches
cb SECS ARM (2) - close (verify)
cb SECS LOGIC (2) - close (verify)
cb EDS (3) - close (verify)
SECS LOGIC (2) - on (up) (verify)
SECS PYRO ARM (2) - on (up) (verify)
EDS PWR - on (up) (verify)
ABORT SYS TWR JETT (2) - on (up) (verify)
NO TWR JETT - continue to orbit

EMERGENCY CSM/LV SEPARATION

COASTING

LV XLUNAR - SAFE
cb SECS ARM (2) - close
SECS LOGIC (2) - ON
SECS PYRO ARM (2) - ARM
ROT CONTR PWR DIRECT (2) - MNA/MNB
SC CONT - SCS
BMAG MODE (3) - ATT 1/RATE 2
SCS TVC SERVO PWR 1 - AC1/MNA
2 - AC2/MNB
(Continue through thrusting)

THRUSTING (TLI)

00:00 TRANS CONTR - CCW (4 Sec)
MN BUS TIE (2) - ON
GMBL MTRS (4) - ON (LMP Confirm)
00:04 TRANS CONTR - NEUTRAL & +X
ΔV THRUST A - NORMAL
THRUST ON PB - PUSH
TRANS CONTR +X - RELEASE
ΔV THRUST (2) - OFF when clear

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FIRE/SMOKE IN CM (CREW SUITED)

WARNING: CM water must not be used to extinguish fire

- 1 CAB FAN (2) - OFF (verify)
- 2 Monitor EPS for excessive current and remove power from affected bus
- 3 Verify suit compressor on good AC bus
- 4 Use fire extinguisher as appropriate

FIRE IS OUT

- 5 Remove smoke from cabin per "Contamination in CM" procedures before removing helmets

FIRE PERSISTS - DUMP CABIN

- 6 Verify:
 SUIT CKT RET vlv - CLOSE (push)
 EMER CAB PRESS vlv - OFF
 PLSS 02 vlv - OFF
- 7 Visually check suit integrity
- 8 CAB PRESS REL (RH) - DUMP to 3.0 psia
 then to BOOST ENTRY
 REMARK: Provides controlled cabin dump until
 suit circuit pressure is verified
- 9 Verify Suit pressure > 3.5 psia
- 10 CAB PRESS REL (RH) - DUMP
 and/or CAB PRESS DUMP vlv - OPEN
- 11 CAB PRESS ind 0.0 psia for 6 min
- 12 CAB PRESS REL (RH) - NORMAL
- 13 CAB PRESS DUMP vlv - CLOSE
- 14 Do not repress cabin until fire source is removed

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FIRE/SMOKE IN CM (CREW UNSUITED)

- 1 CAB FAN (2) - OFF (verify)
- 2 SUIT COMPR (2) - OFF
- 3 Monitor EPS for excessive current and
remove power from affected bus
- 4 Don emergency O2 masks
- 5 Use fire extinguishers as appropriate

FIRE IS OUT

- 6 Remove smoke from cabin per "Contamination in CM"
procedure before removing O2 masks

FIRE PERSISTS - DON SUITS and DUMP CABIN

- 7 Don PGA's except helmets and verify O2 connectors
(Use O2 masks as long as possible)
- 8 DIRECT O2 vlv - OPEN
REMARK: Purges suit circuit of smoke and fumes
- 9 Don helmet
- 10 SUIT FLOW vlv (3) - SUIT FULL FLOW
- 11 SUIT COMPR 1 (2) - AC1 (AC2)
- 12 DIRECT O2 vlv - CLOSE
- 13 EMER CAB PRESS vlv - OFF
- 14 Visually check suit integrity
- 15 CAB PRESS REL (RH) - DUMP to 3.0 psia
then to BOOST ENTRY

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- 16 Verify Suit pressure holding >3.5 psia
- 17 CAB PRESS REL (RH) - DUMP
and/or CAB PRESS DUMP vlv - OPEN
- 18 CAB PRESS ind 0.0 psia for 6 min.
- 19 CAB PRESS REL (RH) - NORMAL
- 20 CAB PRESS DUMP vlv - CLOSE
- 21 Do not repress cabin until fire source is removed

FIRE/SMOKE IN CM DURING ENTRY

- 1 CABIN FAN (2) - OFF (verify)
- 2 Monitor EPS indicators for excessive current.
Immediately remove power from affected bus.
- 3 ROT CONTR PWR DIRECT (2) - MNA/MNB
& maintain attitude if required.
- 4 If affected bus is:

MNA

AC INV 1 AC BUS 1 - OFF
AC INV 2 AC BUS 1 - on (up)
Set up for CM/RCS sys 2
AUTO RCS SEL A/C ROLL (4) - OFF
AUTO RCS SEL CM 1(6) - OFF
AUTO RCS SEL CM 2(6) - MNB
RCS dump is fuel rich

MNB:

AC INV 2 AC BUS 2 - OFF
AC INV 1 AC BUS 2 - on (up)
RCS dump is oxidizer rich

- 5 CAB PRESS REL vlv (RH) - DUMP
- 6 Continue ENTRY

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LAUNCH EMERGENCY POWERDOWN

(MN BUS Voltage <26.0, no short verified
Powerdown until MN BUS voltage >26.5 vdc)

	<u>Amps</u>
O2 HTRS (2) - off (ctr)	11.00
If GET > 2 min	
EDS AUTO - OFF	
2 ENG OUT - OFF	
LV RATES - OFF	
cb MNA BAT C - close	
cb MNB BAT C - close	
S BD PWR AMP - off (ctr)	3.53
FC PUMPS (3) - OFF	12.20
H2 HTRS (2) - off (ctr)	1.44
LIGHTS (min req'd)	
TAPE RCDR FWD - off (ctr)	1.82
VHF AM B - off (ctr)	1.00
*GMBL MTRS P2, Y2 - OFF	10.00
IMU PWR ON (STBY)	5.70
CMC MODE - FREE	
G&N IMU PWR - OFF	
CMC to STBY	2.00
V48E	
F 04 46, load 0 in left digit R1	
PRO,PRO,PRO,V46E	
V37E 06E	
F 50 25 00062 CMC PWR DN	
PRO, hold until STBY lt on	
G&N PWR - OFF	1.50
ECS GLY PUMP 1 - OFF	2.77
cb ECS RAD CONT/HTR (2) - open	2.69
PWR SCE - off (ctr)	0.65
TELECOM GRP 1&2 - OFF	2.20
cb INSTR ESS MNA&B - open	4.70

*Crew Option

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BUS LOST RECONFIGURATION

A. Loss of MN BUS A

EDS AUTO - OFF

FC 2 - MNB only

FC 1 (MNA&B) - off (ctr)

cb MNB BAT C - closed

cb MNA BAT BUS A - open

INV 3 - MNB, AC1

SCS TVC PITCH & YAW - RATE CMD

FDAI SEL - 2

TVC GMBL DRIVE PITCH & YAW - 2

BMAG MODE (3) - RATE 2

RHC PWR DIR (2) - MNB

AUTO RCS PITCH, YAW, & B/D ROLL (12) - MNB

Insertion Checklist

FC 1 - MNB (if req'd to maintain MNB voltage)

SUIT H2O ACCUM - AUTO 2

RAD FLOW AUTO CONT - 2

B. Loss of MN BUS B

EDS AUTO - OFF

FC 2 - MNA only

FC 3 (MNA&B) - off (ctr)

cb MNA BAT C - closed

cb MNB BAT BUS B - open

INV 3 - MNA, AC2

TVC GMBL DRIVE PITCH & YAW - 1

RHC PWR DIR 1 - MNA

AUTO RCS PITCH, YAW, & B/D ROLL (12) - MNA

Insertion Checklist

FC 3 - MNA (if req'd to maintain MNA voltage)

ECS RAD HTR - PRIM 2

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C. Loss of BAT BUS A

EDS AUTO - OFF

cb MNA BAT C - closed

TVC GMBL DRIVE PITCH & YAW - 2 (if bus lost
prior to GMBL MTR turn on)

AUTO RCS PITCH, YAW, & B/D ROLL (12) - MNB

Insertion Checklist

After attempting to restore bus

MN BUS TIE BAT B/C - on (up)

cb MNA BAT C - open

cb MNA BAT BUS A - open

cb MNB BAT C - open

cb MNB BAT BUS B - open

For subsequent main bus ties

cb MNA BAT C - closed

cb MNB BAT BUS B - closed

cb PYRO A SEQ A - open

ECS RAD HTR - PRIM 2

D. Loss of BAT BUS B

EDS AUTO - OFF

cb MNB BAT C - closed

TVC GMBL DRIVE PITCH & YAW - 1 (if bus lost
prior to GMBL MTR turn on)

AUTO RCS PITCH, YAW, & B/D ROLL (12) - MNA

Insertion Checklist

MN BUS TIE BAT A/C - on (up)

cb MNA BAT C - open

cb MNA BAT BUS A - open

cb MNB BAT C - open

cb MNB BAT BUS B - open

For subsequent main bus ties

cb MNB BAT C - closed

cb MNA BAT BUS A - closed

cb PYRO B SEQ B - open

ECS RAD HTR - PRIM 1

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E. Loss of AC BUS 1

AC INV 1 MNA - OFF
SUIT COMPR 2 - AC2
FDAI SEL - 2
BMAG MODE (3) - RATE 2
TVC SERVO PWR 1 - AC2/MNB (loss of ϕ A only)
SCS TVC (P&Y) - RATE CMD

Insertion Checklist

S BD NORM PWR AMP - SEC
FC PUMP 1 - AC2
ECS GLY PUMP - AC2
BMAG 1 PWR - OFF
G&N PWR - AC2
SIG COND/BIAS PWR 1 - AC2
Maintain GLY EVAP TEMP INLET temp above 40°F
ECS RAD FLOW CONT - 2

F. Loss of AC BUS 2

AC INV 2 MNB - OFF
TVC SERVO PWR 2 - AC1/MNA (loss of ϕ A only)
SCS TVC (P&Y) - AUTO
 Δ V CG - LM/CSM
Control SPS with trim wheels
S BD NORM XPNDR - PRIM

Insertion Checklist

FC PUMP (2&3) - AC1
BMAG 2 PWR - OFF
FDAI SEL - 1
SIG COND/BIAS PWR 2 - AC1
Activate SEC COOL LOOP
Shut down PRIM EVAP

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LOSS OF TWO FUEL CELLS AT LIFTOFF

Entry Area 3-1

After 2 min EDS AUTO - OFF

1. If loss of FC 1&2, tie BAT C to MNA
2. If loss of FC 2&3, tie BAT C to MNB
3. If loss of FC 1&3, tie BAT C to both MNA&B

Failed FC PUMPS (2) - OFF

At insertion perform insertion checklist plus the following power down. Leave batteries on until power down complete.

Pnl 2

02 & H2 HTRS (4) - off (ctr)
C/W NORM - ACK

Power down CMC per checklist, leave IMU powered up (Note: No need to realign IMU for 3-1 entry).

Pnl 3

TAPE RCDR FWD - off (ctr)
S BD NORM PWR AMP - off (ctr)
Remaining fuel cell to both Main Busses. Select single inverter operation.

Pnl 5

cb RAD HTR OVLD (2) - open

Pnl 7

LOGIC 2/3 PWR - OFF
FDAI/GPI PWR - OFF
ELEC PWR - OFF
BMAG PWR (2) - ON

Note: If voltage permits leave BMAG PWR in WARMUP after the batteries are off the main bus, if not place BMAG PWR - ON 40 min prior to IMU/GDC align.

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Pnl 8

AUTO RCS SEL (16) - OFF
cb SCS LOGIC (4) - open

4. Batteries off line
Charge lowest battery
5. 1 hour to SPS ignition
BMAG PWR (2) - WARMUP

Pressurize CM RCS
6. 10 min to SPS ignition
Power up CMC and SCS and perform IMU/GDC align
7. Perform SPS normal deorbit

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CSM 106

(00:00 to 00:42) (10K)

00:00 TRANS CONTR - CCW then NEUTRAL
CM/SM SEP (2) - on (up)

00:14 ELS - AUTO
ELS LOGIC - on
TWR JETT (2) - on (up)
APEX COVER JETT PB - PUSH

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00:16 DROGUE DEPLOY PB - PUSH
      CM RCS He DUMP PB - PUSH
      Monitor altimeter
      If <3800 ft - DEPLOY MAINS
      >3800 ft - NO ACTION
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00:28 If <10,000 ft - DEPLOY MAINS

GO TO LANDING PHASE AT 10,000 ft pg L/6-7

(00:42 to 16.5 nm) (1:56)

00:00 TRANS CONTR - CCW then NEUTRAL
CM/SM SEP (both) - ON

00:11 CANARD DEPLOY - PUSH

```
00:14  ELS - AUTO
        ELS LOGIC - on (up)
        RCS CMD - ON
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GO TO LANDING PHASE pg L/6-6

MODE IC ABORT

(16.5 nm to TWR JETT) (01:56 - 03:16)

00:00 TRANS CONTR - CCW then NEUTRAL
CM/SM SEP (2) - on (up)
RCS CMD - ON

00:11 CANARDS DEPLOY
CM RCS PRESS - on (up)
RCS TRNFR - CM
RCS IND - CM (1 or 2)

S/C PLATFORM GO/NO GO (Excessive Rates)
V82E Check HA

HA>32nm & PLAT GO

HA<32nm or PLAT NO GO

TWR JETT sw(2)-on(up)
MAN PITCH - RATE CMD
ENT ATT RO°, P135°, Y0°
BMAG (3)- ATT1/RATE 2
EMS FUNC - ENTRY
EMS MODE - NORMAL
At .05G Lt,
.05G sw - on (up)
Fly Max Lift

Estab. +5°/SEC
pitch rate
EXCESSIVE + PITCH RATES
*ROLL 90° *
*USE YAW THRUSTERS TO *
*CONTROL RATE *
ROLL BACK TO HEADS DN

θ (.05G)
GET DROGUE

LET FAILS To JET - L/5-3

GO TO LANDING PHASE pg L/6-6

Basic Date April 18, 1969
Changed

MODE II RCS ABORT

(TWR JETT to MODE III) (03:16 - 09:41)

00:00 TRANS CONTR - CCW (4 sec min) & HOLD
No BECO-Reset THC,Req. RSO Shutdown
*Reset & start DET *

00:03 *CSM/LV SEP - PUSH*
*RCS CMD - ON *

00:05 TRANS CONTR - NEUTRAL THEN +X

00:24 TRANS CONTR +X OFF
Entry ATT - (R=0°,P=120°,Y=0°)(Comp1 by 1:40)
V82E - NOTE TFF

If TFF>2 min, Yaw 45° (LEFT) out-of-plane

BMAG MODE (3) - ATT1/RATE 2

cb MNA&B BAT C (2) - closed

CM/SM SEP - on (up) GET 400K

CSM/LM FNL SEP (2) - on (up)

CM RCS - PRESS 0 (.05G)

RCS TRNFR - CM GET DROGUE

C&W MODE - CM

EMS FUNC - ENTRY

EMS MODE - NORMAL

Set up Single Ring RCS

At .05G Lt, Sw - on (up)

EMS ROLL - ON

Fly Max. Lift

N62E

GO TO LANDING PHASE pg L/6-6

Basic Date April 18, 1969
Changed

Basic Date April 18, 1969
Changed

CSM 106

GO TO INSERTION (+11:53) pg L/3-3

LANDING PHASE (30K, DESCENDING)

30K' cb ELS (2) - close

ELS - AUTO

ELS LOGIC - on (up)

24K' Twr jett (auto)

*TWR JETT (2) - on (up) *

*CSM/LM FNL SEP(2)-on(up) *

Apex cover jett (auto)

*APEX COVER JETT PB-PUSH) *

(WAIT 2 SECS)

Drogues deployed (auto)

DROG DPLY PB-PUSH

If Drogues Fail:

*ELS - Man *

*STABILIZE CM *

5K' MAIN DPLY PB - PUSH

*ELS - AUTO *

23.5K' Cabin Pressure increasing

*If not increasing by 17K': *

*CABIN PRESS REL vlv (RH)-DUMP *

Basic Date — April 18, 1969

Changed —

CSM

MODE IA

10K'

Main parachutes deployed

MAIN DEPLOY PB - PUSH (within 1 sec)

VHF ANT - RECY

VHF AM A - SIMPLEX

VHF BCN - ON

RCS DUMP

CABIN PRESS REL vlv (2) - CLOSE

DIRECT 02 vlv - OPEN

CM RCS LOGIC - on (up)

CM PRPLNT - DUMP (burn audible)

MONITOR CM RCS 1&2 for He press decrease

NO BURN or PRESS DECREASE

* USE BOTH RHC's *

*DO NOT FIRE PITCH JETS *

CM PRPLNT - PURGE (to zero He press)

CM RCS He DUMP PB-PUSH

*RHC (both) - 30 secs *

* NO PITCH *

CABIN PRESS REL vlv - BOOST/ENTRY

STRUT LOCKS UNLOCK

cb FLT & PL BAT BUS A,B,&BAT C (3) - close

cb FLT & PLT MNA & B (2) - open

cb ECS RAD HTR OVLD (2) - open

cb SPS P&Y (4) - open

3K'

CABIN PRESS REL vlv (RH) - DUMP

FLOOD Lts - POST LDG

CM RCS PRPLNT (2) - OFF

ROT CONTR PWR DIRECT - OFF

800'

CAB PRESS REL vlv - CLOSE (latch off)

MN BUS TIE (2) - OFF

AFTER LANDING:

cb MAIN REL PYRO (2) - close

MAIN REL - on (up)

GO TO POST LANDING pg E/7-3

Basic Date — April 18, 1969
Changed —

CSM 106

TLI PREPARATION

GET = 1:50 Don Helmets & Gloves
SUIT CKT RET vlv - CLOSE (push)

XLUNAR - INJECT (verify)
EDS PWR - on (verify)

CYI LOS EMS AV Test pg G/1-45

(01:55:23)

SUNSET

(01:59:41)

TAN AOS (V)

(02:09:07)

SECS PYRO ARM (2) - on (up)

TRANS CONTROL PWR - ON

ROT CONTR PWR NORMAL (2) - AC/DC

(verify)

ROT CONTR PWR DIRECT (2) - MNA/MNB

LV IND/GPI - SII/SIVB (verify)

LV GUID - IU (verify)

cb DIRECT ULLAGE (2) - closed

Set EVENT TIMER to 51:00

Begin MONITOR For TB6

TAN LOS

(02:15:58) V48E 31102,01111

ARIA AOS KEY V83E

(__:__:__) SET ORDEAL

(2:23:48) TB 6 - SII SEP Lt on (TIG-9 min, 38 sec)

SII SEP Lt out (38 sec later)

51:00

Start DET COUNTING UP

SC CONT - SC5 (verify)

MONITOR LV TANK PRESS

CRO AOS

AP < 36 psid (OXID > FUEL)

(02:25:22)

AP < 26 psid (FUEL > OXID)

EMERGENCY CSM/LV SEP pg L/5-3

Lox TK Press > 50 psia Fuel < 25

OX < 20

Inhibit SIVB Restart

MER AOS

UP TLM CM - BLOCK (verify)

(02:28:49)

UP TLM IU - BLOCK (verify)

TLI PREP

Basic Date April 18, 1969
Changed 2:3, 1969

CSM 106

57:00 V37E 47E (check bias) Record _____
 (Limit : 9.8 fps/min)
 F 16 83 AVX,Y,Z (.1fps)
 ORDEAL - 200/LUNAR
 ORDEAL FDAI 1 - ORB RATE
 FDAI 2 - INERTIAL
 SLEW FDAI to PITCH = 17°

58:00 N62E
 F 16 62 VI,HDOT,HPAD (fps,fps,.1nm)
 MONITOR VI () at ECO
 TVC SERVO PWR 1 - AC1/MNA
 2 - OFF
 TAPE RCDR - FWD/RCD/HBR/CMD RESET
 EMS MODE - NORMAL

58:20
 CRO LOS
 (02:31:36)

58:36 SII SEP Lt - ON
 TLI Inhibit Signals will not
 * be honored after 59:42 *

58:38 SIVB ULLAGE Begins

59:00 SLEW FDAI to PITCH = 9.0°

59:42 SII SEP Lt - off (TIG - 18 sec)

59:52 SIVB FUEL LEAD

59:55 SIVB ULLAGE discontinues
 Insure FDAI 1 PITCH = 5.0°

59:59 Start ORDEAL Torquing
 LV ENG 1 Lt - on

00:00 SIVB IGNITION (: : GETI)

00:02 LV ENG 1 Lt - off

RED AOS
 (02:34:00)

MER LOS
 (02:35:30)

SUNRISE
 (02:38:00) MONITOR THRUST & ATTITUDE
 MONITOR LV TANK PRESS
 SIVB ECO (Lt on) (BEGIN TB7)
 *EMER SIVB CUTOFF AT 6 SEC *
 * PAST BURN TIME IF VI ATTAINED*
 *LV STAGE sw - SII/SIVB *
 *If still no ECO, *
 *THC CCW & NEUTRAL IN 1 SEC *
 (MSFN AOS) UP TLM - ACCEPT
 (02:43:43) ~~XXXX~~

KEY VERB (freeze display)

TLI PREP

Basic Date Jul 18, 1969
 Changed May 3, 1969

CSM 106

L
7-3

SIVB ATT HOLD 20 sec & BEGIN VENTING
SIVB MNVR TO ORB RT (HDS DN) (.3°/sec)

Record VI _____
HDOT _____
HPAD _____

KEY RLSE

F 16 62

KEY RLSE

F 16 83

ΔVX,Y,Z

(.1fps)

TVC SERVO PWR 1 - OFF

PCM BIT RATE - LOW

EMS MODE - STBY

EMS FUNC - OFF

SECS PYRO ARM (2) - SAFE

FDAI 1 - INERTIAL

PRO

Report VI _____
ΔV _____

F 37

OOE *(When CMC ACTY Lt - out)*
ΔV66E
CMP To LH Couch
CDR To CTR Couch

02:50:00 Report radiation survey meter
reading and ~~GET~~ GET of reading

Basic Date April 18, 1969
Changed May 1, 1969

CSM 106

TLI 10 MIN ABORT

SECS LOGIC (2) - on (up)(verify)
SECS PYRO ARM (2) - ARM

00:00 TRANS CONTR - CCW (4 sec)
DET RESET (verify)
00:03 SIVB/CSM SEP
LV ENG 1 Lt - out
* CSM/LV SEP PB - PUSH*
* RCS CMD - ON *
00:05 TRANS CONTR - neutral then +X for
10 sec
SIVB/GPI sw - GPI
00:14 TRANS CONTR +X - OFF
V37E 00E
PITCH UP to LOCAL VERT (+X axis
toward the earth)
RATE - LOW
BMAG MODE (3) - ATT1/RATE 2
EDS PWR - OFF
SECS PYRO ARM (2) - SAFE
SECS LOGIC (2) - OFF
cb SECS ARM (2) - open
cb EDS (3) - open
01:00 TRANS CONTR -X (8 to 10 sec)
RATE - HIGH

MNVR TO RETRO ATT
R _____ (180°)
P _____ () 267°
Y _____ (0°)

Basic Date April 18, 1969
Changed May 2, 1969

CSM 106

TLI, LOI ABORTS

L
8-2

RETRO UPDATE

GETI	_____	0 .05G	_____
		GET DROGUE	_____
ΔV	_____	ENTRY P	_____
VC	_____	R	_____
Δtb	_____	Y	_____
GET 400K	_____		

ALIGN HORIZ ON RET +1° MK
GMBL CHECK (Time Permitting)
MN BUS TIE (2) - ON
GMBL MTRS (4) - ON (LMP Confirm)

RATE - LOW
EMS MODE - STBY
EMS FUNC - ΔV SET/VHF RNG
SET ΔV from chart
EMS FUNC - ΔV
EMS MODE - NORMAL

09:45 ΔV THRUST A - NORMAL
V37E 47E (THRUST MONITOR)

F 16 83 ΔVX,Y,Z (.1 fps)

NOTE: For aborts during 1st min of TLI,
KEY V82E F 16 44 (Ha,Hp,Tff)
Burn until Hp < 19NM.

09:50 TRANS CONTR +X

10:00 THRUST ON PB - PUSH
TRANS CONTR +X - OFF
BURN ΔV req'd
ΔV THRUST (2) - OFF
Report cutoff

GMBL MTRS (4) - OFF (LMP Confirm)
TRANS CONT PWR - OFF
TVC SERVO PWR (2) - OFF
MN BUS TIE (2) - OFF
cb SPS P1&2, Y1&2 - open

PRO
00E

F 37

TLI, LOI ABORTS

Basic Date April 18, 1969
Changed May 3, 1969

CSM 10

Go to ENTRY PREP & SUPERCIRC ENTRY PROCEDURE
If est. time to EI<01:55:00 omit MCC and
enter the SUPERCIRC CKLIST as early as possible.
If est. time to EI>01:55:00 anticipate a
MCC. Enter the ENTRY PREP CKLIST at step 9
pg E/1-1

TLI 90 MIN ABORT

TLI+25 Normal CSM/LV Separation- If decision
 to abort made before TLI+25 min,
 abort at this time. If abort deci-
 sion occurs after separation start
 with V37E 00E.
 SECS LOGIC (2) - on (up)
 MSFN Confirm GO for PYRO ARM
 SECS PYRO ARM (2) - ARM

00:00 TRANS CONTR - CCW (4 sec)
 DET RESET (verify)

00:03 SIVB/CSM SEP
 LV ENG 1 Lt - out
 CSM/LV SEP PB - PUSH
 *RCS CMD - ON *

00:05 TRANS CONTR - neutral then +X for
 10 sec
 SIVB/GPI sw - GPI

Basic Date April 18, 1969
Changed May 3, 1969

CSM 106

L
8-4

00:14

TRANS CONTR +X - OFF
V37E 00E
PITCH UP to LOCAL VERT (+X axis
toward the earth)
RATE - LOW
BMAG MODE (3) - ATT1/RATE 2
EDS PWR - OFF
SECS PYRO ARM (2) - SAFE
SECS LOGIC (2) - OFF
cb SECS ARM (2) - open
cb EDS (3) - open

01:00

TRANS CONTR ^{-X} (8 to 10 sec)

RATE - HIGH

MNVR TO RETRO ATT

R _____ (Block Data)

P _____ (Block Data)

Y _____ (Block Data)

RETRO UPDATE (NO COMM - use Block Data)

GETI _____ 0 .05G _____

ΔV _____

GET DROGUE _____

VC _____

ENTRY P _____

Δtb _____

R _____

GET 400K _____

Y _____

XX:XX

Set DET counting up to GETI
ALIGN HORIZ ON RET +1° MK
GMBL CHECK (Time Permitting)
MN BUS TIE (2) - ON
GMBL MTRS (4) - ON (LMP Confirm)

RATE - LOW

EMS MODE - STBY

EMS FUNC - ΔV SET/VHF RNG

SET ΔV from chart

EMS FUNC - ΔV

EMS MODE - NORMAL

TAPE RCDR - CMD RESET/HBR/RCD/FWD

Basic Date April 18, 1969
Changed May 3, 1969

CSM 106

L
8-6

LOI MODE I ABORT

ABORT - Stop Clock
ΔV Thrust A/B - OFF
SPS INJ vlvs (4) - CLOSED
SPS He tb (2) - bp
GMBL MOTS (4) - OFF (LMP Verify)
TVC SERVO PWR 1&2 - OFF
SC CONT - SCS
PCM BIT RATE - LOW
EMS MODE - STBY
RECORD DATA AND COMPUTE PAD

F 97 40

Record TFC 59 59
Vg _____
Vm _____
EMS Vc _____

PRIMARY

G&N ΔVm _____
VcABORT(Chart) _____
GET LOI1 _____
+15:00

ENTR

F 99 40

Maneuver to LOI1 Attitude

ENTR

F 16 85

Record Vx _____
Vy _____
Vz _____

GET 15 min ABORT : :

PRO

F37 V82E

ALTERNATE
ΔVcLOI1 PAD _____
EMS Vc(Shutdown) _____
Vc(Burned) _____
Vc ABORT(Chart) _____

F 16 44

Record H _____
Hp _____
TFF _____

PRO

F 37 00E

V66E

Basic Date April 18, 1969
Changed May 3, 1969

CSM 106

MAN ATT PITCH - ACCEL CMD
SET in GDC THUMBWHEELS - (176.8, 68.0, 6.2)
MAN MNVR To ABORT R,P,Y
GDC ALIGN
CHECK DAP (P&Y TRIM)
EMS FUNC - ΔV SET/VHF RNG
SET ΔV_c ABORT
EMS FUNC - ΔV

TVC CHECK & PREP

cb STAB CONT SYS (all) - close
cb SPS (12) - close
RATE - LOW (verify)
LIMIT CYCLE - ON
MAN ATT (3) - RATE CMD
BMAG MODE (3) - ATT1/RATE2
ROT CONTR PWR DIR (2) - OFF
SCS TVC (2) - AUTO
TVC GMBL DRIVE P&Y - AUTO
MN BUS TIE (2) - ON (verify)
TVC SERVO PWR 1 - AC1/MNA
TVC SERVO PWR 2 - AC2/MNB
TRANS CONTR PWR - ON
ROT CONTR PWR NORMAL #2 - AC
RHC #2 - ARMED

- 5 m

PRIMARY TVC CHECK

GMBL MOT P1, Y1 - START/ON (LMP Verify)
THC - CW
Verify NO MTVC

SEC TVC CHECK

GMBL MOT P2, Y2 - START/ON (LMP Verify)
SET GPI TRIM
Verify MTVC
THC - NEUTRAL
Verify GPI Returns To Trim Pos
ROT CONT PWR NORM (2) - AC/DC
ATT DB - MIN (verify)
ROT CONTR PWR DIR (2) - MNA/B
SPS HE vlvs (2) - AUTO, verify tb-bp
LIMIT CYCLE - OFF
FDAI SCALE - 50/15 (verify)

Basic Date April 18, 1969
Changed May 1, 1969

L
8-8

-2m

EMS MODE - NORMAL
V37E 47E

Δ V THRUST A(B) - NORMAL

THC - ARMED (verify)

RHC (2) - ARMED

PCM BIT RATE - HIGH

-10s

ULLAGE

00

THRUST ON PB - PUSH

SPS THRUST Lt - on

MONITOR THRUSTING

Pc 95-105 psia

EMS COUNTING DOWN

SPS INJ vlvs (4) - OPEN

SPS HE vlvs tb (2) - gray

SPS FUEL/OXID PRESS - 175 to 195 psi

ECO

Δ V THRUST (2) - OFF

SPS INJ vlvs (4) - CLOSED

SPS He tb (2) - bp

GMBL MOTS (4) - OFF (LMP Verify)

TVC SERVO PWR 1&2 - OFF

MN BUS TIE (2) - OFF

PCM BIT RATE - LOW

BMAG MODE (3) - RATE 2

F 16 83

RECORD Vx _____

Vy _____

Vz _____

PRO

F37 00E

V66E

EMS Δ Vc _____

Basic Date April 18, 1969
Changed July 3, 1969

CSM 106

TLI BACKUP GUIDANCE PROCEDURES

TB5

KEY V46E
GUIDANCE - CMC
ORDEAL ALT = 200 NM
EARTH/LUNAR = LUNAR
SET THUMBWHEELS
R = 180°, P = 116°, Y = 0° (1st OPP)
R = 180°, P = 115°, Y = 0° (2nd OPP)
SELECT FDAI #1
FDAI SOURCE - ATT SET /IMU
MANEUVER LV TO THUMBWHEELS ATT
(ZERO NEEDLES)
KEY V16 N20E (DSKY R2 = 116.00°)
NULL LV RATES
ORDEAL FDAI #1 - ORB RATE
ORDEAL MODE - FAST/HOLD
SLEW FDAI #1 TO P = 0°
INSURE R2 = 116.00°

TB6

51:00 S-II SEP LT - OFF
START DET COUNTING UP FROM 51:00
57:00 INSURE FDAI #1 P = 0° & R2 = 116.00°
57:40 ORDEAL MODE - OPERATE/SLOW
(STARTS ORDEAL TORQUING)
MANEUVER LV TO
P = 0°, Y = +3.5° AND MAINTAIN UNTIL BURN COMP
58:00 KEY V37E 47E
N20E
58:36 S-II SEP LT - ON
58:38 LV ULLAGE START
59:42 S-II SEP LT - OFF (IGN - 18 SEC)
59:50 N62E
59:52 SIVB FUEL LEAD
59:55 LV ULLAGE TERMINATES
59:59 ENG #1 LT - ON
00:00 IGNITION
00:02 ENG #1 LT -OFF
FLY P = 0°, Y = 3.5°
05:44(1st) CUTOFF ON PAD VI THC - CCW FOR 1 SEC & RESET
05:29(2nd)

Basic Date April 18, 1969
Changed May 9, 1969

CS: 06

CSM 106 Basic Date

SC CONT - CMC/FREE
COAS PWR - ON
AUTO RCS SEL (16) - MNA/MNB
TVC SERVO PWR 1 - AC1/MNA
TVC SERVO PWR 2 - OFF
cb DOCK PROBE (2) - close (verify)
WASTE STOW vlv - CLOSE
DIRECT 02 vlv - OPEN till CAB PRESS
ind - ~5.7 psia, then CLOSE

(02:54:00) SIVB MNVRS to SEP ATT and goes inertial
BMAG MODE (3) - RATE 2
MAN ATT (3) - MIN IMP
ATT DB - MIN
RATE - LOW
SC CONT - SCS *cb SECS ARM (2) - close (verify)*
cb SECS LOGIC (2) - closed (verify)
RCS CMD - ON
Check RCS (all quads) (audible)
SC CONT - CMC/FREE
MAN ATT (3) - RATE CMD

(03:00:00)SECS LOGIC (2) - on (up) (verify)

MSFN confirm GO for PYRO ARM, SIVB
venting stopped, and GO for T&D
V48E 11102, 01111

V46E

V66E

V16N20E

V16N22E

EMS FUNC - ΔV SET/VHF RNG
Slew ΔV ind to -100.0

V62E

TAPE RCDR - CMD RESET/HBR

2 SEPARATION

V37E 47E

SECS PYRO ARM (2) - on (up)
EMS FUNC - ΔV
EMS MODE - NORMAL
SC CONT - CMC/AUTO

DET START

THC - +X (audible)
CSM/LV SEP PB - PUSH
LV ENG 1 Lt - out
LV TK PRESS ind (4) - 0
THC - +X to 0.8 fps

*NO SEP:

*THC - CCW

*DET - RESET and counting (auto)

LV ENG 1 Lt - out, LV TK PRESS ind - 0

*THC - Neutral at 0.8 fps

SM RCS PRPLNT tb (8) - gray

(00:35) THC -X to 0.3 fps

3 TRANSDPOSITION

V62E

MAN ATT PITCH & YAW - ACCEL CMD
Pitch up 180° at 2°/sec

MAN ATT PITCH & YAW - RATE CMD

V37E 00E

V49E (SC AUTO MNVR TO DOCKING ATT)

EDS PWR - OFF

cb EDS (3) - open

H1 Gain Ant Activation

cb HGA FLT BUS - close

cb HGA GROUP 2 - close

HGA TRACK - MAN

HGA PWR - on (up)

Slew ant to verify operation

Acquire HGA phaselock: $P = -53^\circ$ $Y = +39^\circ$

For TV operation go to pg S/1-21

When TV operation is completed:

HGA TRACK - MAN

Slew antenna to $PITCH = -52^\circ$, $YAW = 270^\circ$

Basic Date - APR 18, 1969
Channel - May 3, 1969

4 DOCKING

PROBE EXTD/REL - RETRACT (tb - gray)

5 CONTACT & CAPTURE

At capture:

PROBE EXTD/RETR tb - bp

(A)(B)

SC CONT - CMC/FREE

Allow probe to damp S/C motions (10 sec)

BMAG MODE (3) - RATE 2 (verify)

When within 2° of docking attitude (min poss)

PROBE RETRACT PRIM - 1

(F)

PROBE EXTD/RETR tb - gray (A-latches 1,5,9:

B-latches 3,7,11)

SECS PYRO ARM (2) - SAFE

SECS LOGIC (2) - OFF

PROBE EXTD/REL - OFF

PROBE RETRACT PRIM - OFF

cb DOCK PROBE (2) - open

PCM BIT RATE - LOW

6 POST DOCKING

RATE - HIGH

TVC SERVO PWR 1 - OFF

COAS PWR - OFF

cb RCS LOGIC (2) - open

EMS MODE - STBY

EMS FUNC - OFF

COUCHES: CDR - 90°, CMP - 0°, LMP - 180°

LM PWR - OFF (verify)

TUNNEL LIGHTS - ON

Basic Date April 18, 1969
Changed April 3, 1969

7 CSM/LM PRESSURE EQUALIZATION (Decal)

02 PRESS IND sw - SURGE TANK

Verify CRYO 02 PRESS 1 ind - 865-935 psia

EMER CAB PRESS sel - OFF (verify)

PLSS 02 vlv - OFF

DIRECT 02 vlv - CLOSE (verify)

TUNL VENT vlv - LM/CM ΔP

LM/CM ΔP ind - +4 psid (pegged)

CRYO 02 PRESS ind - 850 psia

PRESS EQUAL vlv - OPEN

CAB PRESS ind - 4.5 psia

PRESS EQUAL vlv - CLOSE

LM/CM ΔP ind - ~2.4 psid

Monitor LM/CM ΔP ind for 3 min

and verify ΔP stable

PRESS EQUAL vlv - OPEN

CAB PRESS ind - 4.0 psia

REPRESS 02 vlv - OPEN

CAB PRESS ind 5.7 psia

Cycle REPRESS 02 as required

between 4.0 and 5.7 psia limits

until REPRESS 02 PRESS ind

~0.0 psig

REPRESS 02 - CLOSE

CAB PRESS ind $\geq$ 4.0 psia

*If CAB PRESS ind < 4.0 psia *

* PRESS EQUAL vlv - CLOSE *

LM/CM ΔP ind - ~0.0 psid

CAB PRESS ind - 5.0 psia

PLSS 02 vlv - FILL

EMER CAB PRESS sel - BOTH

TUNL VENT vlv - OFF

WASTE STOW vlv - VENT (until cabin purge complete
at ~8 hrs)

Basic Date - April 18, 1969
Changed - May 3, 1969

- 8 TUNNEL HATCH REMOVAL (Decal)
PRESS EQUAL vlv - open (CCW) (verify)
ACTR HNDL SEL - unstow, pull to stop, set to U
- push to stop
Verify gearbox disconnect socket - U
ACTR HNDL SEL - stow
- push to stow
Remove hatch, stow

- 9 DOCKING LATCH VERIFICATION (Decal)
LATCH HNDL - Pull to verify hook engaged
Not engaged - attempt to engage
* before recocking *
- LATCH IND Button (red) - flush
Power Bungee Fairing - parallel to +X
(push +X end of Bungee before recocking
if not parallel)
*Unlocked latches: *
- * Recock latches *
- * *Hook does not release *
- * *AUX REL (yellow) - push*
- * *Cock latch *
- *Release latch - push man release *
- GN2 BLEED button (red) - press (10 sec)

- 10 LM UMBILICAL CONNECTION (Decal)
LM Connector Fairings (2) (orange) - open
LM Umbilical Connectors (2) - install and lock
LM Connector Fairings (2) (orange) - close
SYS TEST - 4D
LM PWR - CSM
SYS TEST ind - 0.5-3.2 volts

- 11 HATCH INSTALLATION (Decal)
Align hatch in tunnel
ACTR HNDL SEL - unstow, set to L
- push to stop
Verify gearbox disconnect socket - L
*If latches cannot be closed: *
- * GEARBOX DISCONNECT - 180° CCW (tool B)*
- * AUX LATCH DRIVE - LATCH (113° CW) *
- * Verify hatch latched, remove tool B *

Basic Date April 18, 1969
Changed 3, 1969

CSM 106

L
9-6

ACTR HNDL. SEL - stow
 - push to stow
PRESS EQUAL vlv - CLOSED (CW)
TUNL VENT - LM/CM ΔP
TUNNEL LIGHTS - OFF

(C)

12 ~~Report Radiation Dosimeter Reading~~
~~GET of Reading~~

13 LM EXTRACTION

LOAD DAP: 21101 V46E
EMS FUNC - ΔV SET/VHF RNG
Slew ΔV ind to -100.0
EMS FUNC - ΔV
EMS MODE - NORMAL

V6 N22 _____, _____, _____

cb SIVB/LM SEP (2) - close (verify)

TVC SERVO PWR 1 - AC1/MNA
SECS LOGIC (2) - on (up)
MSFN Confirm GO for PYRO ARM & Withdrawal
SECS PYRO ARM (2) - on (up)

V66E

V37E 47E

(00:00) SIVB/LM SEP - on (up)

Start DET

(00:05)THC -X for 3 sec

(00:14) CMC MODE - AUTO

V37E 00E

V49E

V62E

SECS PYRO ARM (2) - SAFE
SECS LOGIC (2) - OFF
cb SECS ARM (2) - open
TVC SERVO PWR 1 - OFF
cb SIVB/LM SEP (2) - open
EMS MODE - STBY

Go to Thrusting Proc pg G/1-31 for SPS Evasive Mnr

Basic Date April 18, 1969
Changed May 1969

CS-006

P30 MANEUVER

SET STARS

R ALIGN _____

P ALIGN _____

Y ALIGN _____

ULLAGE _____

HORIZON/WINDOW _____

OTHER _____

PURPOSE
PROP/GUID

WT	N47

P TRIM N48

Y TRIM

HRS GETI

MIN N33

SEC

ΔV_X N81

 ΔV_Y ΔV_Z

R

P

Y

H_A N44

HP

 ΔVT

BT

 ΔVC

SXTS

SET

TDN

BSS

SDA

0148

DATE

LAT NO 1

LONG

RTGU EMS

VIC

GET 0.05G

Basic Date ~~April 18, 1969~~
Changed ~~May 3, 1969~~

CSM 106

LM INTERFACE

1 IVT TO LM FOR FIRST ENTRY

Couches: CDR - 0°, CMP - 0°, LMP - 180°

TUNL LTS - ON

TUNL VENT vlv - LM/CM ΔP

Verify LM/CM ΔP <0.2

*LM/CM ΔP >0.2 *

Equalize CM/TUNL pressure

* (Decal) *

Remove hatch & stow (Decal) (3)

Remove probe & stow (Decal) (4)

Remove drogue & stow (Decal) (5)

Read docking tunnel index angle

Open LM hatch

LMP Transfer to LM (6)

At LM request,

LM PWR - RESET, then OFF

SYS TEST - 4D

SYS TEST ind - 0 volts

Perform comm checks with LM

At LM request

LM PWR - CSM

SYS TEST - 4D

SYS TEST ind - 0.5 - 3.2 volts

LMP Transfer to CSM (6A)

Close LM hatch

Install CM hatch (Decal) (11)

TUNL VENT vlv - LM PRESS

TUNL LTS - OFF

2 IVT TO LM FOR SECOND ENTRY

Couches: CDR- 0°, CMP- 0°, LMP- 180°

Don helmet protective shield (if req'd)

Suit Integrity Ck (if req'd)

TUNL LTS - ON

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TUNL VENT vlv - LM/CM ΔP

Verify LM/CM $\Delta P < 0.2$

*LM/CM $\Delta P > 0.2$ *

* Equalize CM/TUNL Pressure*

* (Decal) *

Remove tunnel hatch (Decal) (3)

Verify docking tunnel index angle

Open LM hatch

LMP transfer to LM (6)

At LM request,

LM PWR - RESET, then OFF

SYS TEST - 4D

SYS TEST ind - 0 volts

CDR transfer to LM (6)

LMP transfer to CSM (6A)

LMP don LCG & PGA

LMP transfer to LM (6)

Remove LM umbilicals (7)

Install drogue (Decal) (8)

Install probe (Decal) (9)

Preload probe (Decal) (10)

LM hatch closed

Verify CSM roll cmds inhibited
until LM/CM $\Delta P > 3.5$ psid

Cock docking latches (12) (G) (H)

Verify hook is clear of LM ring

*Hook does not release: *

* AUX REL (yellow) - push*

* Cock latch *

Install tunnel hatch (Decal) (11)

Perform hatch integrity check (Decal) (12)

Remove center couch and stow

Install docking target

DOCKING TARGET - BRIGHT

Receive target alignment verification from LM

Configure side hatch for EVT

ACTR HANDLE SEL - N (neutral)

GN2 VLV HANDLE - pull (inboard)

GN2 PRESS ind - minimum

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3 TUNNEL HATCH REMOVAL (Decal)

PRESS EQUAL vlv - open (CCW) (D)
ACTR HNDL - unstow, pull to stop, set to U (D)
- push to stop
Verify gearbox disconnect socket - U
ACTR HNDL SEL - stow
- push to stow
Remove hatch, stow

4 PROBE REMOVAL (CM Side) (Decal)

A. Translunar Docking:

Verify EXTEND LATCH engaged indicator
(red) not visible

*EXTEND LATCH not engaged: *

* PRELOAD SEL LEVER - rotate CW (away from*	*
* orange stripe)	*
* PRELOAD HANDLE - Torque CCW to engage	*
* extend latch (red ind. not visible)	*

GN2 BLEED button (RED) - press (10 sec)

PRELOAD SEL LEVER - rotate CCW (parallel)
to orange stripe)

PRELOAD HNDL - Torque (CW) unload support beams

B. Lunar Orbit Docking:

PRELOAD SEL LEVER - rotate CW(away from orange stripe)

PRELOAD HNDL - torque CCW to engage EXTEND LATCH (red
indicator not visible)

GN2 BLEED button (red) - press (10 sec)

Basic Date April 18, 1969
Changed April 29, 1969

C. Both TLD & LOD:

PROBE UMBILICALS (2) (yellow) - disconnect and stow
Elec connector covers (2) (yellow) - close
PRELOAD HNDL - position against umbilical connector
PRELOAD SEL LEVER - mid position
INSTALLATION STRUT - unstow, position on tunnel
wall (yellow marks)
CAPTURE LATCH RLSE HNDL LOCK - Rotate CCW to unlock
(orange stripe visible)
RATCHET HNDL - unstow to full extension
- push to first detent (red band)
- push outbd and hold to fold
probe ()
RATCHET HNDL - pull to full extension
- ratchet one stroke only
Restow RATCHET HNDL and INSTALLATION STRUT
CAPTURE LATCH RLSE HNDL - Pull, rotate to unlock
(180° CW) - push to recess

*Capture latches will not release: *

* Ratchet probe forward *

* Preload probe until latches release*

Remove PROBE - pull aft to release (25 lbs)

5 DROGUE REMOVAL (Decal)

LOCK LEVER - Pull, rotate 90° CCW
DROGUE - rotate CW, push clear of support
- remove from tunnel

6 CREW TRANSFER TO LM

CDR and LMP Audio Panels:

PWR - OFF

SUIT PWR - OFF

AUDIO CONT - NORM

CDR and LMP SUIT FLOW vlv - OFF

Connect to TRANSFER UMB if desired

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Changed - April 29, 1969

6A CREW TRANSFER TO CSM

CDR and LMP Audio Panels:

Verify/set PWR - OFF

Verify/set SUIT PWR - OFF

Verify/set AUDIO CONT - NORM

Verify/set CDR and LMP SUIT FLOW vlv - OFF

Connect to TRANSFER UMB if desired

LMP transfer to CSM

7 REMOVE LM UMBILICALS (FINAL)

LM Connector Fairings (2) (orange) - open

Connectors (2) - release and remove

Fairings (2) - close

Pull lanyard on LM end of umbilical

Remove umbilicals from tunnel, stow in F1 or F2

8 INSTALL DROGUE (Decal)

DROGUE - Align Lugs with fittings

- Rotate CCW to stops

LOCK LEVER - Rotate 90° CW to detent

9 INSTALL PROBE (Decal)

CAPTURE LATCH RLSE HNDL - Pull, rotate CCW to cock
pos (150°)

Push PROBE into DROGUE

CAPTURE LATCH RLSE HNDL - rotate CCW to LOCK position
(do not force)

- push to recess

Verify capture latches engaged (CDR)

INSTALLATION STRUT - unstow, position on tunnel
wall (yellow marks)

RATCHET HNDL - unstow to full extension (green band)

- ratchet probe fwd to orange hash mark (k)

Restow RATCHET HNDL and INSTALLATION STRUT

Verify RATCHET PAWL indicator (red) flush with housing

*Ratchet pawl indicator not flush: *

* Hold RATCHET HANDLE full outboard *

* Press Pawl indicator to seat (flush)*

* Release RATCHET HANDLE *

Basic Date April 18, 1969
Changed April 29, 1969

CSM 10

Preload Shaft - push up into detent
CAPTURE LATCH RLSE HNDL - Set in detent
CAPTURE LATCH RLSE HNDL LOCK - Rotate CW to lock
(orange stripe not visible)
PROBE UMBILICALS (2) (yellow) - connect to dock ring

10 PRELOAD PROBE (Decal)

PRELOAD SEL LEVER - rotate CCW (parallel to orange stripe)
PRELOAD HNDL - torque (CW) to release (F)
Verify capture latches engaged (CDR)
PRELOAD HNDL - Push inboard to detent
- pos 45° to support beam
PRELOAD SEL LEVER - mid position
Verify CAPTURE LATCH RLSE HNDL LOCK is locked
(orange stripe not visible)

11 HATCH INSTALLATION (Decal)

Align Hatch in tunnel
ACTR HNDL SEL - unstow, set to L
- push to stop
Verify gearbox disconnect socket - L

*If latches cannot be closed: *
GEARBOX DISCONNECT - 180° CCW (tool B)
*AUX LATCH DRIVE - LATCH (113° CW) *
*Verify hatch latched, remove tool B *
*(Cannot remove hatch from LM side) *

ACTR HNDL SEL - stow
- push to stow
PRESS EQUAL vlv - CLOSED (CW) (C)

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12 HATCH INTEGRITY CHECK (Decal)

Verify LM Hatch Closed, DUMP vlv - AUTO (CDR)
Verify CABIN PRESS ind - 4.7-5.3 psi
TUNL VENT vlv - TUNL VENT for 30 sec
- LM/CM ΔP , check ΔP
- Recycle to TUNL VENT until $\Delta P > 3.5$

*Cannot vent tunnel: *
* If O2 FLOW ind. increases, open hatch, *
* wipe seal surfaces, close hatch *
* If O2 FLOW ind. does not increase, dump *
* tunnel through LM during reg. check *
* Monitor LM/CM ΔP & flow to ck integrity *

Verify LM/CM ΔP ind constant ($\pm .2$) at last value
for 2 min

Verify O2 FLOW ind - no increase

Before Undocking only:

TUNL VENT vlv - LM TUNL VENT for 10 min, then to
LM/CM ΔP

Verify LM/CM $\Delta P > 4.0$ (pegged)

TUNL VENT vlv - OFF

TUNNEL LIGHTS - OFF

MALFUNCTION LIST

TUNNEL HATCH

A. Positive Indication of No Capture

- THC -X withdraw to formation
flight distance
- PROBE EXTD/REL - EXTD/REL for 5 sec
- RETR
- PROBE EXTD/REL tb (2) - gray (verify)
- Attempt redocking as before

B. One tb does not indicate bp but capture attained

- For retraction, use bottle (1) in system with
gray tb
- If no retraction, use bottle (2) in system with
gray tb
- If no retraction, use bottle (1) in system with
bp tb
- Trouble shoot later before removing probe as
follows:

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Changed - May 7, 1969

CSM 10

- DOCK PROBE RETRACT (2) - OFF
- cb DOCK PROBE (2) - open
- Interchange probe umbilical connections
(cut cable retainers if necessary)
- Cock docking latches #1 and #7
- cb DOCK PROBE (2) - closed
- DOCK PROBE EXTD/REL - RETR
- If previous bp tb is now gray,
failure is in probe. Interchange
umbilicals again. Only one bottle
is usable to complete mission
- If previous bp tb is now bp, failure
is in SC wiring. Two bottles are usable
to complete mission as connected.
- Manually release docking latches
#1 and #7

C. Pressure Equalization Valve Will
Not Close.

- Remove Hatch
- Use Tool B In External Tool Interface For Additional
Leverage

- D. Pressure Equalization Valve Will Not Open For TLD:
- Vent CM
 - Perform Tunnel Operations
 - Repress CM

For Subsequent IVT

TUNL VENT vlv - LM PRESS

(May require up to 12 hrs
to equalize pressure)

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Changed - May 7, 1969

PROBE

- E. Do not get retraction using PRIM-1
 - Initiate retraction using bottles in the following order:
 - PROBE RETRACT SEC-1
 - If no retraction, initiate PROBE RETRACT (2) - 2 simultaneously
- F. Preload Ratchet Will Not Drive To Achieve Proper Preload.
 - Use Tool F To Drive Hex Fitting On Aft End Of Preload Shaft (CW Direction, 30 To 40 lbs On Tool Handle)

DOCKING LATCHES

- G. Cannot Cock Docking Latch By Pulling Handle
 - Depress Aft End Of RH No-Back Pawl While Pulling On Latch Handle.
 - If unsuccessful, Use Tool E to depress LH No-Back Pawl while pulling on Latch Handle

TUNNEL

- H. High O2 Flow While Cocking Docking Latches
 - Re-engage/verify 3 latches ~ 120° apart are engaged
 - Use Tool F to drive hex fitting on aft end of preload shaft 180° CW
 - Torque preload handle CW until load limiter releases
 - Disengage docking latches

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Changed - May 7, 1969

SIDE HATCH

- I. Cannot latch side hatch (frozen gearbox)
 - The Following tools are required:
 - tool B, tool F, (3) jackscrews
 - Install (3) jackscrews to restrain hatch in closed position
 - Use tool B to remove (2) clevis pins connecting linkage to gearbox and (1) clevis pin from linkage in corner above gearbox.
 - Tighten jackscrews to close hatch as far as possible
 - Use tool F on flats of latch bellcrank to drive latch to over-center closed position (Apply tool F to upper latch on hinge side to drive the lower and hinge side linkage closed. Apply tool F to center latch to drive upper linkage closed. Gearbox side linkage may not close if gearbox is in full open position.
 - Install (2) clevis pins in threaded holes in linkage bell cranks at upper gearbox side and lower hinge side. (Clevis pins installed when approx. half the threads are visible).

PROBE

- J. Both tb's not gray after undocking
 - PROBE EXTD/REL - EXTD/REL for 5 sec
 - PROBE EXTD/REL - RETR
 - PROBE EXTD/REL tb (2) - gray (verify)
- K. Pushing ratchet handle outboard does not ratchet probe forward
 - Push ratchet handle to first detent (red band)
 - Slowly push ratchet hndl outboard ~ 25° until audible click. (If pushed outboard past point of click, probe will release).
 - Repeat until orange hash mark is visible.

Basic Date April 18, 1969
Changed MAY 7, 1969

Changed _____

- CSM 106

LM JETTISON

- 1 Remove Probe And Drogue, Stow In LM (4) (5)
- 2 CDR and LMP Transfer TO CSM (6A)
- 3 Close LM Hatch
- 4 POO, V49, Load LM Jett attitude
~~FOG 22~~ CMC MODE - AUTO
PRO BMAG MODE (3) - RATE 2
PRO To Maneuver
- 5 Install tunnel hatch (Decal) (11)
Perform hatch integrity check (Decal) (12)
- 6 cb SECS ARM (2) - CLOSE
 SECS LOGIC (2) - on (up)
 Obtain GO from MSFN
 SECS PYRO ARM (2) - ARM
- 7 At Jett Attitude:
 BMAG MODE (3) - ATT 1/RATE 2
 S/C CONT - SCS
 Load DAP N46: 11102
- 8 CSM/LM FINAL SEP(2) - ON (.4 fps sep)
 SECS PYRO ARM (2) - SAFE
 SECS LOGIC (2) - OFF
 cb SECS ARM (2) - open
 MNVR to SEP Attitude
- 9 Sep: P47
 EMS MODE - NORMAL
 Sep 4 jet - X 2.0 fps

Basic Date April 18, 1969
Changed May 3, 1969

EARTH ORBIT BLOCK DATA

X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C
X	X				X	X				AREA
X	X	X			X	X	X			LAT
X	X				X	X				LONG
										GETI
X	X	X			X	X	X			ΔV_C

REMARKS:

[illegible]

**E.O.
BLOCK**

Basic Date 24, 1969
Changed _____

CONTINGENCY EVA

A CM PREP FOR CONTINGENCY EVA

- 1 PGA Bag Stowed
- 2 EVA Stabilizer Strut installed
- 3 Center Couch removed and stowed under L Couch
- 4 C AND R SUIT FLOW vlvs are OFF, VERIFY INTERCONNECTS
INSTALLED
- 5 GN2 VLV HANDLE - PULL
- 6 REMOVE PIP PIN, STOW IN R-3
- 7 L and R Couch - 270 degree
- 8 Jack Screws fully opened, tethered, and accessible
- 9 Tool B tethered and accessible
- 10 Unstow CM INGRESS/EGRESS BAR
- 11 SUIT FLOW vlv - CABIN FLOW
- 12 SUIT RETURN vlv - OPEN (pull)
- 13 EMER CAB PRESS SEL - BOTH
- 14 ECS HOSES - RED/RED, BLUE/BLUE

B SYSTEM PREPARATION FOR DEPRESS

- 1 COMM: SIMPLEX A, VERIFY CM/LM COMM
- 2 O2 PRESS IND sw - SURGE TANK
- 3 CRYO TK O2 PRESS 1 IND - 865-935 psi
- 4 Verify REPRESS O2 PRESS - 865-935 psig
- 5 Select Attitude Control Mode and
Manuever Spacecraft to EVT Attitude
- 6 AUTO RCS SELECT -
UNDOCKED
A/C ROLL - A1, A2 - OFF
PITCH - A3, - OFF
YAW - B3 - OFF
DOCKED
ALL - OFF

C PLSS/OPS COMM CHECK (ON CDR REQUEST)

- 1 VHF AM A - DUPLEX
- 2 Verify VOICE COMM with LMP
(NO SIMUL RCV A&B, SIMP A-CDR,DUP A - LMP)

Basic Date April 18, 1969
Changed April 3, 1969

CSM 106

D FINAL SYSTEMS PREP FOR DEPRESS

- 1 Stow loose items
- 2 EXT APU/EVA LT - on (up) (if req'd)
- 3 EXT PMSZ LT - off (ctr)
- 4 Verify PGA FLOW DIVERTER vlv - HORIZONTAL
- 5 Verify Feedport Cover - Locked
- 6 Don Helmet and Shield
- 7 Don Gloves
- 8 Secure Helmet Stowage Bag
- 9 SUIT RETURN vlv - CLOSE (push)
- 10 EMERG CABIN PRESS SEL - OFF
- 11 Verify helmet, gloves, zipper, and hoses locked

E SUIT CKT INTEGRITY CHECK (Decal)

DIRECT O2 vlv - CLOSE
SUIT PRESS ind - 4.7-5.3 psia
O2 FLOW ind - 0.2-0.4 lb/hr

CAUTION

SUIT TEST vlv should remain
in the PRESS position until
suit circuit pressure is sta-
bilized to preclude seal scarring

If repositioning of SUIT TEST
vlv from PRESS is required
prior to suit pressure and O2
flow stabilization, perform
the following:

- a. O2 DEMAND REG vlv - OFF
- b. Allow 15 sec (min)
stabilization time
- c. Reposition SUIT TEST
vlv - DEPRESS or OFF
as applicable
- d. When suit pressure
stabilized, O2 DEMAND
REG vlv - BOTH

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Changed April 23, 1969

SUIT TEST vlv - PRESS
02 FLOW ind - 1.0 lb/hr (pegged)
02 FLOW HI lt - on
M/A - ON, Reset
Cycle SUIT CKT RTN AIR vlv OPEN
and CLOSE at suit press
of 1.5 to 2.0 psig
SUIT PRESS ind - 8.8-9.8 psia
PGA PRESS ind - 4.1-4.5 psig
02 FLOW HI lt - out
Allow 02 flow to stabilize 15 sec
02 flow shall remain below
0.8 lb/hr for 30 secs after
stabilization
SUIT TEST vlv - DEPRESS
02 FLOW ind - 0.2-0.4 lb/hr
SUIT PRESS - slightly > CAB PRESS
SUIT TEST vlv - OFF
02 DEMAND REG vlv - BOTH (verify)

F CABIN DEPRESS (Decal)

- 1 Confirm GO for Cabin Depress with CDR
- 2 Verify CABIN FAN (2) - OFF
- 3 PLSS 02 vlv - FILL
- 4 REPRESS 02 vlv - CLOSE
- 5 Verify CAB PRESS REL vlv (2) - NORMAL
(SAFETY LATCH ON)
- 6 SIDE HATCH DUMP vlv - OPEN
(02 FLOW HI Warning Light may come on
prior to Cabin Press Reg Lock-Up)
- 7 Monitor Cabin Pressure to 3.25 psia
- 8 At 3.25 psia, SIDE HATCH DUMP vlv - CLOSE
- 9 Verify 02 FLOW IND - < 0.5 LB/HR
- 10 Verify Cabin Pressure at 3.25 psia and CM Suit
Circuit Pressure Stable at 3.5-4.0 psia
- 11 SIDE HATCH DUMP vlv - OPEN
- 12 CABIN PRESS ind - 0.0 psia
- 13 SIDE HATCH DUMP vlv - CLOSE

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Changed April 3, 1969

CSM 106

G HATCH OPENING (Decal)

- 1 GN2 VLV HANDLE - PULL (verify)
Gauge reads min
- 2 LOCK PIN RELEASE KNOB - UNLOCK
- 3 Verify Lock Pin Indicator Released
- 4 Verify GEAR BOX SEL - UNLATCH
- 5 Verify ACTR HANDLE SEL - U
- 6 Unstow ACTR HANDLE
- 7 Unlock hatch slowly
- 8 Verify hatch unlocked
- 9 GEAR BOX SEL - LATCH
- 10 ACTR HANDLE SEL - L
- 11 Stow ACTR HANDLE
- 12 Open hatch to the full open position
- 13 CMP give GO for Transfer
- 14 Mark CDR OPS activation & record

H CONTINGENCY INGRESS

- 1 CDR Ingress To LEB
- 2 Secure Position, Manage Lifeline
- 3 CMP Stow CM INGRESS/EGRESS BAR
Before LMP Ingress
- 4 LMP Ingress CM, Center Couch Area
- 5 PLSS FEEDWATER vlv - CLOSED (up)
- 6 Connect R Electrical Umbilical
and verify COMM

I HATCH CLOSING (Decal)

- 1 Close hatch
- 2 Lock Hatch & verify Lock Pin dropped in
- 3 ACTR HANDLE SEL - N
- 4 Stow ACTR HANDLE
- 5 GEAR BOX SEL - LATCH (verify)

J CDR VAC TRANSFER TO CM ECS

If 20 minutes elapsed from initial OPS O2 opening,
connect CDR to ECS

- | | |
|--|---|
| *Verify SUIT FLOW - OFF | * |
| *Connect R O2 Umbilicals | * |
| *PURGE vlv - CLOSE | * |
| *SUIT FLOW vlv - FULL FLOW, verify flow* | * |
| *OPS O2 SHUTOFF - CLOSE | * |

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Changed May 3, 1969

K CABIN REPRESS to 3.0 PSIA (Decal)

- 1 REPRESS 02 vlv - OPEN For 10 SEC Then CLOSE
(CABIN PRESS APPROX 1.0 PSIA)
- 2 CABIN PRESS ind - MONITOR FOR 30 SEC (Gross Leakage)
- 3 REPRESS 02 vlv - OPEN
- 4 CONTROL SURGE TANK PRESS GREATER THAN
150 PSIA)
- 5 REPRESS 02 - CLOSE
- 6 CABIN PRESS IND - 3.0 PSIA
- 7 DUMP OPS INTO CABIN IF AVAIL

L REPRESS FROM 3.0 PSIA TO 4.8 PSIA If OPS NOT AVAIL

- 1 CABIN REPRESS vlv - OPEN
- 2 MAINTAIN 150 PSIG MIN IN SURGE TK
- 3 CAB PRESS 4.7-5.3 psia
- 4 02 PRESS IND sw - TK 1
- 5 CAB REPRESS vlv - OFF
- 6 Go to Final Systems Config, Step Q.

M OPS REPRESS

STEPS N, O, P MUST BE ACCOMPLISHED

N CDR CONN TO ECS

- 1 VERIFY R SUIT FLOW - OFF
- 2 CONNECT R 02 UMB
- 3 OPS 02 vlv - CLOSE
- 4 DEPRESS USING PURGE VLV
- 5 SUIT FLOW vlv - FULL FLOW (verify flow)
- 6 PURGE vlv - CLOSE

O LMP CONN TO ECS

- 1 VERIFY C SUIT FLOW - OFF
- 2 PLSS 02 - OFF (up)
- 3 DEPRESSURIZE USING PURGE VLV
- 4 REMOVE OPS 02 HOSE & PURGE VLV
- 5 CONNECT C 02 UMB
- 6 SUIT FLOW vlv - FULL FLOW (verify flow)
- 7 PLSS PUMP - OFF
- 8 FAN - OFF

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Changed 7 3, 1969

CSM 106

P DISCONNECT OPS 02 HOSE AND HOLD

- 1 OPS 02 vlv - OPEN
- 2 CAB PRESS 4.7-5.3 PSIA
- 3 OPS 02 vlv - CLOSE

Q FINAL SYSTEMS CONFIGURATION

- 1 CABIN PRESS ind - 4.7-5.3 psia
- 2 PLSS 02 vlv - FILL
- 3 VERIFY REPRESS 02 INCREASING
- 4 EMER CABIN PRESS SEL - BOTH
- 5 SUIT RET vlv - OPEN (pull)
 - DOFF Gloves and Temp Stow
 - DOFF Helmets and EV VISORS and Temp Stow
 - Verify SURGE TANK PRESS
 - 865-935 psi
 - 02 PRESS IND sw - TK 1
 - EXT LTS - RUN/EVA - OFF

R PLSS/OPS DOFFING

1 OPS -

- Verify Antenna Stowed
- Verify OPS 02 Shutoff vlv - CLOSE
- Stow OPS 02 Hose
- Remove OPS Straps from PGA

2 PLSS/OPS -

- Stow PLSS Antenna
- Remove RCU from PGA
- Disconnect 02 and H2O Hoses
- DOFF PLSS
- Hold PLSS in Position for Restowage

WARNING

- Before Disconnecting RCU from PLSS,
- Verify
 - PUMP - OFF (down)
 - FAN - OFF
 - MODE SEL - POS 3

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Changed April 3, 1969

- 3 Disconnect RCU and Stow in Helmet Stowage Bag
- Stow O2 Plugs in PGA
- Stow PLSS O2 and H2O Hoses
- Stow PLSS COMM Umbilical
- Secure OPS O2 Actuator
- Stow OPS O2 Hose
- Temp Stow PLSS/OPS and OPS

S PREP FOR EQUIPMENT JETTISON

- 1 COMM CARRIER Donned
- 2 O2 Hoses Connected to PGA
- 3 Protective Plugs Installed In Left
PGA O2 Connectors
- 4 Stow Loose Items
- 5 Prepare all Equipment to Be Jettisoned
and Secure -
PLSS/OPS
OPS
EV VISORS (2)
Place RCU, PURGE vlv (2), OPS/PGA Straps,
CM ECS Interconnects, Lifeline and Waist
Tethers in Helmet Stow Bag

T PREP FOR DEPRESS: GO TO-(A)-AND COMPLETE REQ'D STEPS

U EQUIPMENT JETTISON

- Jettison Equipment -
- PLSS/OPS, Helmet Stowage Bag,
 - OPS
 - EV VISORS (2)

V FINAL CABIN CONFIGURATION

- 1 Reinstall Center Couch
- 2 Reinstall PGA Stowage Bag
- 3 Remove EVA Stabilizer Strut
- 4 Restow Tool B and Jack Screws
- 5 Install Hatch Pip Pin

Basic Date April 18, 1969
Changed 3, 1969

CSM 1106